COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 9 : SLO – 2 | Logical expressions and circuits
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Part B – Multiple Choice Questions

- 1. C — All inputs are 1
- 2. B — OR
- 3. B — 1
- 4. B — (A·B)' (AND followed by NOT = NAND)
- 5. C — 4 rows (2^2)

Part A – Match the Following

- 1. AND gate — c) Multiplication of inputs (A·B)
- 2. OR gate — a) Output is 1 if at least one input is 1
- 3. NOT gate — b) Output is the opposite of the input
- 4. Truth Table — d) Tabular representation of inputs and outputs
- 5. Logic Circuit — e) Diagram showing gates connected together

Part C – Activity Questions

- 1. Truth table for F = A·B + C (8 rows): 000→0, 001→1, 010→0, 011→1, 100→0, 101→1, 110→1, 111→1.
- 2. Circuit: inputs A and B feed a 2-input AND gate; the AND output and input C feed a 2-input OR gate; OR output = F.
- 3. Real-life example — car seatbelt warning: buzzer = (ignition ON) AND (seatbelt NOT fastened) OR (door OPEN). Uses AND (ignition·belt■), NOT (belt status inverted), and OR (adds door condition) together.